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AND TECHNOLOGY**

DEPARTMENT OF ELECTRONIC AND COMPUTER ENGINEERING

BSC ELECTRONIC AND COMPUTER ENGINEERING

PROJECT PROPOSAL

**DESIGN AND IMPLEMENTATION OF A LOW-COST
MULTICHANNEL**

**SURFACE ELECTROMYOGRAPHY CONTROL INTERFACE
FOR UPPER-LIMB PROSTHETIC REHABILITATION**

Submitted by:

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*A Final Year Project Proposal submitted to the Department of Electronic and
Computer Engineering in partial fulfillment of the requirements for the award of a
Bachelor of Science Degree in Electronic and Computer Engineering.*

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DECLARATION

This project proposal is my original work, except where due acknowledgement is made in the text, and to the best of my knowledge has not been previously submitted to Jomo Kenyatta University of Agriculture and Technology or any other institution for the Award of a degree or diploma.

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TITLE OF PROJECT:

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LIMB PROSTHETIC REHABILITATION**

SUPERVISOR CONFIRMATION:

This project proposal has been submitted to the Department of Electronic and Computer Engineering, Jomo Kenyatta University of Agriculture and Technology, with my approval as the University supervisor:

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ABSTRACT

Access to upper-limb prosthetic technologies in sub-Saharan Africa is constrained by cost, with commercial myoelectric control interfaces priced between USD 3,000 and USD 8,000. This project proposes the design and implementation of a low-cost, four-channel surface electromyography (sEMG) control interface for upper-limb prosthetic rehabilitation. The system captures residual-limb muscle activity through surface electrodes, conditions the signals using a dedicated analog front-end built around INA128 instrumentation amplifiers with Sallen-Key bandpass filtering (20--500 Hz) and a 50 Hz twin-T notch stage, and samples four channels at 2 kHz on an ESP32-S3 microcontroller. Time-domain features are extracted from 250 ms analysis windows and classified into eight hand gestures using a compact 1D convolutional neural network quantized to INT8. The classified gestures drive a 5-finger tendon-driven 3D-printed prosthetic hand actuated by five SG90 servo motors (fingers) and one MG996R servo motor (wrist) via a PCA9685 I2C PWM driver, while a Bluetooth Low Energy (BLE) mobile application provides guided user calibration and live monitoring. The methodology follows a six-phase iterative engineering workflow: simulation, single-channel proof of concept, PCB design and fabrication, ML training and deployment, system integration, and validation with five test subjects. The system targets a signal-to-noise ratio above 20 dB, macro F1 score of 0.85 or higher, inference latency below 50 ms, and an electronics bill of materials below USD 20. The expected results include a validated four-channel acquisition board, a real-time embedded classification pipeline, and a functional prosthetic demonstrator at a fraction of the cost of existing systems.

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LIST OF ABBREVIATIONS

sEMG -- Surface Electromyography

ESP32-S3 -- Espressif Systems Microcontroller with dual-core Xtensa LX7

BLE -- Bluetooth Low Energy

CNN -- 1D Convolutional Neural Network

TFLite -- TensorFlow Lite for Microcontrollers

INA128 -- Instrumentation Amplifier (Texas Instruments)

ADC -- Analog-to-Digital Converter

BOM -- Bill of Materials

PCB -- Printed Circuit Board

DOF -- Degrees of Freedom

SNR -- Signal-to-Noise Ratio

CMRR -- Common-Mode Rejection Ratio

DMA -- Direct Memory Access

PWM -- Pulse Width Modulation

PCA9685 -- 16-channel I2C PWM driver

NinaPro -- Non-Invasive Adaptive Prosthetics (public sEMG dataset)

MAV -- Mean Absolute Value

RMS -- Root Mean Square

WL -- Waveform Length

ZC -- Zero Crossings

SSC -- Slope Sign Changes

F1 -- Harmonic Mean of Precision and Recall

LDA -- Linear Discriminant Analysis

SVM -- Support Vector Machine

RF -- Random Forest

INT8 -- 8-bit Integer Quantization

IEEE -- Institute of Electrical and Electronics Engineers

WHO -- World Health Organization

KES -- Kenyan Shilling

CHAPTER ONE

INTRODUCTION

1.1 Background of the Study

Surface electromyography (sEMG) is a non-invasive technique for observing electrical activity associated with voluntary skeletal-muscle contraction. Motor-unit action potentials produce small voltage variations at the skin surface; after amplification and filtering, these signals can be analysed to infer intended movement. Typical sEMG energy is concentrated approximately within the 20--500 Hz range, with raw amplitudes between 50 microvolts and 5 millivolts, while practical measurements are affected by 50 Hz power-line interference, motion artefacts below 20 Hz, electrode-skin impedance variations, and thermal noise from the acquisition electronics [1], [6].

For myoelectric prosthetic control, surface electrodes positioned over residual-limb musculature capture sEMG signals during voluntary contractions. A signal-conditioning and classification pipeline amplifies the differential signal, suppresses unwanted frequency components, samples the conditioned waveform, computes discriminative features from analysis windows, and maps the resulting feature vector to a hand gesture. The gesture is then translated into actuator commands for a prosthetic mechanism. Multichannel acquisition, typically four to eight channels, provides spatial diversity across different muscle groups, which is essential for distinguishing more than two or three gestures reliably [6], [7].

Commercial myoelectric prostheses from manufacturers such as Ottobock and Ossur achieve classification accuracy exceeding 95 percent but cost between USD 10,000 and USD 40,000. The control interface electronics alone account for USD 3,000 to USD 8,000 of this total. Even research prototypes documented in the literature cost USD 500 to USD 1,500 for the electronics subsystem [4], which is prohibitive in developing economies. Globally, an estimated 30 million people require prosthetic or orthotic devices, with sub-Saharan Africa carrying a disproportionate burden at approximately 0.5 percent population prevalence [10]. In Kenya, the National Council for Persons with Disabilities registers over 30,000 individuals requiring upper-limb assistive devices, yet the Kenyatta National Hospital orthotics

department, the largest referral facility in East and Central Africa, fits fewer than 200 powered prosthetic arms per year.

Three recent developments make a truly low-cost system feasible. First, the ESP32-S3 microcontroller provides dual-core 240 MHz processing, 512 KB SRAM, hardware-accelerated vector operations, BLE 5.0, and TensorFlow Lite for Microcontrollers support for approximately USD 5 [1]. Second, TensorFlow Lite for Microcontrollers enables deployment of quantized neural networks on memory-constrained devices with models under 100 KB. Third, recent publications from 2025 and 2026 have demonstrated that sEMG classification on ESP32 platforms is achievable with inference latency under 10 milliseconds [1], proving the concept's technical viability for real-time prosthetic control.

Four sEMG channels represent a practical engineering trade-off between spatial information, electrode count, hardware complexity, and computational load. Literature consensus indicates that four to eight channels is the optimal range for distinguishing six to eight gestures reliably [6], [7]. Beyond eight channels, classification gains diminish while hardware cost and wiring complexity increase substantially. This project adopts four channels as the minimum that provides sufficient spatial diversity for the target eight-gesture vocabulary while keeping the electronics BOM below USD 20.

1.2 Problem Statement

The electronic control interface for myoelectric prosthetics -- comprising the sensing, signal conditioning, embedded processing, gesture classification, and actuator-control layers between muscle activity and mechanical movement -- remains prohibitively expensive for users in developing economies. Commercial systems price the control electronics at USD 3,000 to USD 8,000, while published research prototypes cost USD 500 to USD 1,500 for the electronics subsystem alone [4].

Within the reviewed literature, low-cost prototypes commonly address only parts of the complete pipeline. Some demonstrate single-channel classification without prosthetic integration [1]. Others achieve high classification accuracy using deep learning but require desktop GPU inference unsuitable for wearable deployment [2]. Systems that do include prosthetic actuation rely on custom ASICs or FPGAs with electronics costs exceeding USD 500 [4]. No published system combines

multichannel sEMG acquisition, on-device CNN classification, user-specific calibration, and prosthetic actuation for under USD 20 in electronics component cost.

This project addresses the following research question: can a complete, four-channel myoelectric control system -- from skin electrode to mechanical grip -- be designed, built, and validated for under USD 20 in component cost using commercially available components and open-source tools?

1.3 Project Justification

Developing a low-cost multichannel sEMG control interface will reduce the electronics cost barrier for myoelectric prosthetic research and rehabilitation prototypes from hundreds of dollars to under USD 20. By moving signal processing and inference onto the embedded device, the system eliminates the need for a continuous PC connection, enabling standalone wearable operation essential for daily-use rehabilitation. A BLE-based calibration workflow allows the system to adapt to individual users' sEMG characteristics in approximately three minutes, addressing the cross-user variability that limits many existing systems. The complete end-to-end demonstration -- from physiological signal acquisition through classification to mechanical actuation -- validates the engineering feasibility rather than evaluating classification in isolation. Additionally, the modular architecture creates a reproducible, open-source platform that other researchers and clinicians can replicate and extend.

1.4 Scope of the Study

This project covers the following: four-channel sEMG acquisition using INA128 instrumentation amplifiers; analog signal conditioning with bandpass filtering and notch rejection; embedded digital signal processing on the ESP32-S3 at 2 kHz per channel; time-domain feature extraction; 1D-CNN gesture classification quantized to INT8; BLE-based user calibration; and actuation of a 5-finger tendon-driven 3D-printed prosthetic hand (five SG90 servo motors for fingers, one MG996R servo motor for wrist, controlled via PCA9685) driven by classified gestures. Validation will be conducted with five test subjects using informed consent under institutional approval.

The following are explicitly outside scope: large-scale deployment or clinical trials; integration with multiple electrode types beyond Ag/AgCl surface electrodes;

frequency-domain or wavelet feature extraction; more than eight gesture classes; certification as a medical device; and use of high-density electrode arrays. The prototype is a research and rehabilitation demonstrator, not a certified clinical prosthesis.

1.5 Objectives

1.5.1 Main Objective

To design, implement, and validate a low-cost, four-channel surface electromyography acquisition and gesture-classification system based on an ESP32-S3 microcontroller, targeting real-time recognition of eight hand gestures with greater than 80 percent accuracy and inference latency below 50 milliseconds, and to demonstrate the classified output through a 5-finger tendon-driven 3D-printed prosthetic hand with an electronics bill of materials below USD 20.

1.5.2 Specific Objectives

1. To design a four-channel analog front-end using INA128 instrumentation amplifiers (gain = 50), Sallen-Key bandpass filtering (20--500 Hz), and a twin-T 50 Hz notch stage, targeting a signal-to-noise ratio above 20 dB and inter-channel crosstalk below -40 dB.
2. To implement a real-time embedded signal processing and gesture classification pipeline on the ESP32-S3, including 2 kHz per channel sampling, 24-element time-domain feature extraction, and deployment of an INT8-quantized 1D-CNN (less than 100 KB) via TensorFlow Lite for Microcontrollers, achieving an offline macro F1 score of 0.85 or higher.
3. To test and evaluate the complete system by integrating the classified gesture output with a PCA9685-driven 5-finger tendon-driven prosthetic hand (5x SG90 finger servos + 1x MG996R wrist servo), validating end-to-end performance with five test subjects, and measuring real-time accuracy, actuation correctness, system latency, and cost against the USD 20 BOM target.

CHAPTER TWO

LITERATURE REVIEW

2.1 Overview

This chapter reviews the technical literature relevant to the proposed system. The review is organised by technical themes rather than individual papers, following the signal processing pipeline from electrode to actuator. Each theme justifies a design choice made in the methodology. The chapter concludes with a critique of existing work, identification of the research gap, and a summary of the proposed contribution.

2.2 sEMG Signal Characteristics, Electrode Placement, and Acquisition

Surface electromyography captures the superposition of motor-unit action potentials through electrodes placed on the skin surface above target muscles. The signal amplitude ranges from 50 microvolts to 5 millivolts with useful frequency content concentrated between 20 Hz and 500 Hz. Practical measurements are contaminated by 50 Hz power-line interference, motion artefacts below 20 Hz, and electrode-skin impedance variations [6]. For forearm-based gesture recognition, differential electrode placement over the flexor carpi radialis, extensor digitorum, flexor digitorum superficialis, and brachioradialis provides spatial diversity across the primary muscle groups involved in hand movements [7].

2.3 Analog Front-End Design: Amplification, Filtering, and Noise Reduction

The analog front-end conditions the raw sEMG signal for digital acquisition. Instrumentation amplifiers such as the INA128 provide high common-mode rejection ratio (CMRR > 120 dB) and low input bias current, making them suitable for differential biopotential measurement. A critical design consideration is gain distribution: concentrating gain at the instrumentation amplifier (e.g., gain = 500) causes saturation from electrode DC offset (approximately 300 mV multiplied by 500 yields 150 V, exceeding supply rails). The correct approach uses moderate first-stage gain (gain = 50) at the instrumentation amplifier, followed by a high-pass filter to remove DC offset, and then a variable post-filter gain stage (4x to 10x) for a total system gain of 200x to 500x [6].

Bandpass filtering (20--500 Hz) is implemented using Sallen-Key second-order active filter topology, which provides unity passband gain with minimal component count. A twin-T passive notch filter at 50 Hz rejects power-line interference with target rejection exceeding 30 dB. For single-supply operation (1.8--6 V), rail-to-rail op-amps such as the MCP6002 (1 MHz GBW) are preferred over dual-supply alternatives such as the TL072 (which requires minimum plus/minus 3.5 V) [6].

2.4 Signal Segmentation, Feature Extraction, and Preprocessing

After digitisation, the continuous sEMG stream is segmented into overlapping analysis windows. A 250 ms window with 50 ms step (80 percent overlap) is the widely accepted standard, balancing classification accuracy against response latency [6]. From each window, time-domain features are computed: Mean Absolute Value (MAV), Root Mean Square (RMS), Waveform Length (WL), Zero Crossings (ZC), Slope Sign Changes (SSC), and Variance. Literature consensus confirms that these six time-domain features compete with frequency-domain and deep-learning features for six to eight gesture classification tasks, while adding negligible computational overhead on microcontrollers [6], [7].

2.5 Machine-Learning Methods for sEMG Gesture Classification

Salgado et al. [1] demonstrated single-channel sEMG hand gesture classification on an ESP32 microcontroller, classifying four gestures with 92 percent accuracy and inference latency under 10 ms. Their work validates that the ESP32 ADC is adequate for sEMG acquisition when properly conditioned, but the single channel severely limits the number of distinguishable gestures. Mekruksavanich et al. [2] proposed a deep residual network with channel attention achieving 97 percent accuracy on NinaPro DB5, but the model requires desktop GPU inference with no embedded deployment path. CNN-based classifiers consistently outperform traditional methods (LDA, SVM, Random Forest) when gesture count exceeds four classes [6]. However, the computational cost of deep networks must be managed through model compression: INT8 quantisation reduces model size by 4x and accelerates inference on integer-only microcontrollers.

2.6 Embedded and TinyML Deployment

TensorFlow Lite for Microcontrollers enables deployment of quantized neural networks on devices with as little as 256 KB of flash and 64 KB of SRAM. The ESP32-S3 provides 512 KB SRAM and 8 MB of PSRAM, with hardware support for 8-bit integer operations. Salgado et al. [1] demonstrated that an ESP32 can perform sEMG inference within 10 ms, leaving substantial margin for the proposed 50 ms end-to-end latency target. The 1D-CNN architecture is selected because it operates on the extracted feature sequence (24 elements per window), not on raw time series, keeping model input compact and inference fast.

2.7 User Calibration and Personalisation

sEMG signals vary substantially across users due to differences in muscle mass, subcutaneous fat, electrode placement, and skin impedance. Lee et al. [3] addressed cross-user variability through federated prototype learning, achieving personalisation with approximately two minutes of calibration data per new user. A key finding is that fine-tuning only the last classification layer via transfer learning is sufficient for user adaptation, which is the approach adopted in this project. The BLE calibration protocol guides the user through eight gestures with three repetitions each in approximately three minutes.

2.8 Prosthetic Actuation and Control Interfaces

Furtado et al. [5] presented a modular low-cost myoelectric prosthetic hand using 2 to 3 degrees of freedom, finding that mechanical simplicity is sufficient for 80 percent of activities of daily living. Molinari et al. [4] demonstrated real-time prosthetic control with 64 channels but required a custom ASIC and FPGA with electronics costs exceeding USD 500. For the proposed system, a PCA9685 I2C PWM driver controls six servo motors -- five SG90 motors actuating individual finger tendons and one MG996R motor for wrist rotation -- on a 5-finger tendon-driven 3D-printed prosthetic hand. Gesture predictions are not sent directly to motors; confidence threshold, temporal debounce, and command hold time reduce unintended transitions from transient classification errors.

2.9 Cost and Accessibility of Existing Implementations

Cost reporting in the literature is inconsistent: many systems claim "low-cost" without specifying what is included or excluded. This project defines three cost tiers

to ensure transparency: Tier A (electronics BOM: ICs, passives, connectors), Tier B (prototype BOM: Tier A plus PCB fabrication, 3D printing, batteries, servos), and Tier C (volume estimate at 100-plus units). The USD 20 target for Tier A will be compared against the electronics costs of Molinari et al. (more than USD 500), Furtado et al. ("low-cost" without quantification), and commercial systems (USD 3,000 to USD 8,000).

2.10 Critique of Existing Literature

The reviewed studies address important components of the myoelectric control problem but reveal critical gaps when evaluated against the requirements of a complete, low-cost, standalone system. Salgado et al. [1] prove ESP32-based inference is feasible but are limited to a single channel and four gestures with no prosthetic integration. Mekruksavanich et al. [2] achieve high classification accuracy but their model requires desktop GPU inference, making it impractical for wearable deployment. Lee et al. [3] address the important problem of cross-user personalisation but do not demonstrate hardware integration or cost analysis. Molinari et al. [4] provide a complete wearable platform with real-time prosthetic control but at an electronics cost exceeding USD 500 using custom ASICs. Furtado et al. [5] demonstrate mechanical integration at claimed low cost but do not report specific BOM figures or on-device ML deployment.

Systematic reviews by Prakash et al. [6] and Abdikenov et al. [7] confirm consensus findings: four to eight channels is optimal for six to eight gesture classification; time-domain features remain competitive with deep-learning features at this gesture count; 250 ms is the accepted analysis window; and CNN-based classifiers outperform SVM and LDA beyond four classes. However, no reviewed system integrates all four critical capabilities -- multichannel acquisition, on-device CNN classification, user personalisation, and prosthetic actuation -- at a component cost below USD 20. Emerging approaches include transformer-based multimodal sEMG recognition [8] and finger position prediction for amputee-centric interaction [9]; however, both are available only as preprints or early publications without embedded deployment paths, and neither addresses cost as a primary constraint.

Table 1: Comparison of Existing sEMG Systems and the Proposed Project

Study	Channels	Acquisition Hardware	Model Type	On-Device Inference	Gesture Classes	Accuracy / F1	Prosthetic Integration	Reported Cost (Electronics)
Salgado et al. [1]	1	ESP32 ADC (direct)	ANN	Yes (ESP32)	4	92%	No	Not reported
Mekruksavanich et al. [2]	16	Myo Armband	CNN-ResBi GRU-SE	No (GPU/desktop)	18	97%	No	N/A
Lee et al. [3]	Multiple	Commercial EMG	Prototype CNN	Edge-capable	8	N/A	No	N/A
Molinari et al. [4]	64	Custom HD-EMG PCB	LDA + FPGA	FPGA	7	>90%	Yes	> USD 500
Furtado et al. [5]	2-3 DOF	Surface EMG (unspecified)	Threshold-based	No (MCU control)	4 basic	Not reported	Yes	"Low-cost" (unquantified)
This project	4	INA128 custom AFE + ESP32-S3 ADC	1D-CNN INT8	Yes (ESP32-S3)	8	target F1 $\geq$ 0.85	Yes	< USD 20 (KES 3,150)

2.11 Proposed Work

To address the identified gaps, the proposed work will design and implement a four-channel sEMG control interface with on-device 1D-CNN classification on an

ESP32-S3 microcontroller, driving a 5-finger tendon-driven 3D-printed prosthetic hand (5x SG90 finger servos + 1x MG996R wrist servo via PCA9685), at an electronics BOM target of USD 20. The system integrates instrumentation amplification, analog bandpass filtering and notch rejection, embedded digital signal processing, INT8-quantized gesture classification, BLE-based user calibration, and servo-driven actuation into a single, battery-powered, untethered platform. The contribution is an integrated engineering implementation and validation of a low-cost end-to-end control interface, not a claim of a new physiological sensing principle. This proposed solution is intended to be affordable, reproducible, and relevant to the Kenyan sociotechnical context, laying the foundation for Chapter Three, which details the methodology for system design, implementation, and evaluation.

CHAPTER THREE

METHODOLOGY

3.1 Overview

This chapter presents the methodology used to design, implement, and evaluate the proposed four-channel sEMG control interface. The methodology is structured according to the three specific objectives: hardware design (Section 3.2), embedded signal processing and machine learning (Section 3.3), and system testing and evaluation (Section 3.4). Each section describes the engineering design decisions, calculations, and validation procedures. The chapter also addresses electrical safety (Section 3.5), cost evaluation (Section 3.6), and the end-to-end demonstration scenario (Section 3.7).

3.2 System Architecture and Hardware Design

The system architecture integrates an analog acquisition section and a digital control section. Four differential electrode pairs feed INA128 instrumentation amplifiers. Each channel passes through a high-pass filter (20 Hz), a low-pass filter (500 Hz), a 50 Hz twin-T notch filter, and an adjustable gain stage. The conditioned channels are sampled by the ESP32-S3 ADC via DMA. The embedded processor performs windowing, feature extraction, and gesture classification before issuing actuator commands through a PCA9685 PWM driver. Figure 1 shows the full system architecture block diagram.

Table 2: sEMG Signal Specification

Parameter	Proposed Specification
Signal type	Surface EMG, differential measurement
Channels	4 (flexor carpi radialis, extensor digitorum, flexor digitorum superficialis, brachioradialis)
Raw amplitude	50 μ V to 5 mV
Frequency band	20--500 Hz with 50 Hz notch rejection
Sampling rate	2 kHz per channel
Analysis window	250 ms (500 samples), 50 ms step, 80% overlap
Features per channel	MAV, RMS, WL, ZC, SSC, Variance (6)
Feature vector	24 elements (6 features x 4 channels)

3.2.1 Analog Front-End Design

The instrumentation amplifier stage uses the INA128 (Texas Instruments) with gain set to 50 by the external resistor R_G . The INA128 gain equation is $G = 1 + 50 \text{ k}\Omega / R_G$, yielding $R_G = 50 \text{ k}\Omega / (50 - 1) = 1.02 \text{ k}\Omega$. At gain 50, a 5 mV sEMG signal produces 250 mV at the output, and the maximum electrode DC offset of approximately 300 mV produces 15 V at the INA128 output, which exceeds the supply rail. Therefore, the high-pass filter following the INA128 removes this DC component before further amplification.

The bandpass filtering is implemented as a cascade of a second-order Sallen-Key high-pass filter (cutoff 20 Hz) and a second-order Sallen-Key low-pass filter (cutoff 500 Hz), using MCP6002 rail-to-rail op-amps. Filter component values are calculated from the standard Sallen-Key design equations with Butterworth coefficients ($Q = 0.707$). For the high-pass stage with $f_c = 20 \text{ Hz}$, selecting $C = 100 \text{ nF}$ yields $R = 1 / (2 * \pi * f_c * C) = 79.6 \text{ k}\Omega$ (nearest standard value: 82 k Ω). For the low-pass stage with $f_c = 500 \text{ Hz}$, selecting $C = 10 \text{ nF}$ yields $R = 1 / (2 * \pi * f_c * C) = 31.8 \text{ k}\Omega$ (nearest standard value: 33 k Ω).

A twin-T passive notch filter at 50 Hz provides power-line rejection with a target depth exceeding 30 dB. The twin-T component values are $R = 33 \text{ k}\Omega$ and $C = 100 \text{ nF}$, yielding $f_{\text{notch}} = 1 / (2 * \pi * R * C) = 48.2 \text{ Hz}$, tuned to 50 Hz by adjusting R to 31.8 k Ω (standard value: 33 k Ω with 5 percent tolerance covers the range). A variable gain stage (4x to 10x) follows the notch filter, providing total system gain of 200x to 500x. The 500 Hz Sallen-Key low-pass filter also serves as an anti-aliasing stage, attenuating spectral components above the 1 kHz Nyquist frequency imposed by the 2 kHz ADC sampling rate. Each channel uses a common signal reference tied to the analog ground mid-rail ($V_{CC}/2$ formed by a resistor divider). A driven right-leg (DRL) circuit is not used at this stage; instead, a single body-reference electrode is connected to the analog ground mid-rail to reduce common-mode interference. The PCB layout separates the analog front-end ground plane from the digital ground plane, with a single-point star connection at the power entry to minimise digital switching noise coupling into the microvolt-level sEMG signal paths.

3.2.2 Electrical Safety

Because surface electrodes are connected to a human participant, electrical safety is addressed explicitly. The system is powered entirely from 9 V batteries with no mains connection, providing inherent isolation from the power grid. The INA128 instrumentation amplifier has high input impedance (10 GOhm) and draws negligible current through the electrodes (less than 1 microampere bias current). Current-limiting resistors (10 kOhm) are placed in series with each electrode lead to limit fault current below 50 microamperes under any single-fault condition. Before any human testing, the system will pass a leakage current test using a multimeter to confirm electrode-to-ground current is below 10 microamperes. Human testing will be conducted only after obtaining institutional ethics approval and written informed consent from each participant.

3.2.3 SNR and Crosstalk Measurement

Signal-to-noise ratio will be measured by injecting a known-amplitude 100 Hz sinusoidal test signal into the differential inputs using a function generator, recording the output on an oscilloscope, and computing $SNR = 20 * \log_{10}(V_{\text{signal}} / V_{\text{noise}})$ where V_{noise} is measured with inputs shorted. The target is $SNR > 20$ dB. Inter-channel crosstalk will be measured by injecting a signal into one channel and measuring the coupled signal on all other channels with inputs grounded. The target is $\text{crosstalk} < -40$ dB.

3.3 Signal Processing and Machine Learning

3.3.1 Data Path

The complete data path from acquisition to classification is: (1) the ESP32-S3 ADC samples four channels at 2 kHz via DMA; (2) samples are collected into a 250 ms sliding window (500 samples per channel) with 50 ms step (80 percent overlap); (3) six time-domain features (MAV, RMS, WL, ZC, SSC, Variance) are computed per channel, producing a 24-element feature vector; (4) the feature vector is normalised using per-channel mean and standard deviation computed during calibration; (5) the normalised vector is fed to the INT8-quantized 1D-CNN for gesture classification.

3.3.2 Model Architecture and Training

The 1D-CNN input is the 24-element feature vector (not raw time series). The model comprises two 1D convolutional layers (16 and 32 filters, kernel size 3), each followed by batch normalisation and ReLU activation, a global average pooling layer, a dense layer (64 units), and a softmax output layer (8 classes). This architecture keeps the model compact while exploiting local correlations between adjacent features within each channel.

Training uses the NinaPro DB5 public dataset (10 subjects, 16 channels at 200 Hz). Four channels are selected (indices 0, 4, 8, 12, evenly spaced around the forearm from the first Myo armband position) to match the proposed hardware configuration. Windowing uses 50 samples at 200 Hz (equivalent to 250 ms), consistent with the hardware window of 500 samples at 2 kHz. The Rest class is undersampled to match the mean gesture class count, addressing the 73 percent Rest-class imbalance observed in the data. The dataset is split into train (70 percent), validation (15 percent), and test (15 percent) sets, with subject independence preserved (no subject appears in more than one split) to avoid data leakage.

Baseline classifiers (LDA, SVM, Random Forest) are trained on the same feature set to establish a performance reference. If all baselines exceed the 0.85 F1 target, the 1D-CNN is still preferred for its embedded deployment path via TFLite Micro. If the baselines match or exceed the CNN, this will be reported as a finding. After training, the 1D-CNN is quantized to INT8 using TensorFlow Lite post-training quantization and benchmarked for model size (target: less than 100 KB), peak RAM usage, and inference time before deployment on the ESP32-S3.

3.3.3 Calibration and Participant Testing

The BLE calibration protocol guides each participant through 8 gestures, 3 repetitions each, with 5 seconds per repetition and 3 seconds rest between gestures, totalling approximately 3 minutes. During calibration, the system collects the participant's sEMG features and uses transfer learning to fine-tune the last dense layer of the pre-trained 1D-CNN to the individual. The personalised model parameters are stored on the ESP32-S3 flash memory.

For the five-subject evaluation, participants will be healthy adults (age 18-35) with no known neuromuscular conditions. Inclusion criteria: ability to perform all 8

gestures with the dominant hand. Electrode placement follows the protocol defined in the signal specification: Ag/AgCl surface electrodes on four forearm muscle bellies, cleaned with alcohol prep. Each participant performs 5 repetitions of each gesture (40 trials total) after calibration. Institutional ethics approval and written informed consent will be obtained before any human data collection. Participants may withdraw at any time without penalty.

3.3.4 Metrics Definition

Accuracy is the proportion of correctly classified gestures out of total trials. Macro F1 is the unweighted average of per-class F1 scores, computed as $F1 = 2 * \text{precision} * \text{recall} / (\text{precision} + \text{recall})$. Actuation correctness is the proportion of classified gestures that produce the correct finger-and-wrist posture on the prosthetic hand (a separate metric from classification accuracy, as decision filtering may reject low-confidence predictions). Calibration improvement is defined as the difference in accuracy between the calibrated and uncalibrated models for each subject.

Latency is distinguished into three components: (1) model inference time, measured from feature vector input to classification output on the ESP32-S3; (2) decision time, including confidence filtering and debounce logic; (3) total system response time, measured from the end of the 250 ms analysis window to servo motion onset. The 50 ms target refers to component (1), model inference time. Total system response time is expected to be 250 ms (window) + 50 ms (inference) + servo latency, which will be reported separately.

3.4 Actuation and End-to-End Integration

The PCA9685 I2C PWM driver generates servo control signals for six servo motors: five SG90 micro-servos actuate individual finger tendons (thumb, index, middle, ring, little -- PCA9685 channels 0--4) and one MG996R servo controls wrist rotation (PCA9685 channel 5) on a 5-finger tendon-driven 3D-printed prosthetic hand based on the open-source Ada Hand design (PETG, 174x193x42 mm palm). The eight classified gestures map to distinct finger-and-wrist postures as follows: Rest (all fingers neutral, wrist centred -- no servo change); Open Hand (all five SG90 fully extended, wrist centred); Power Grasp (all five SG90 fully flexed, wrist centred); Pinch Grip (thumb + index flexed, middle/ring/little extended, wrist centred); Point (index fully extended, all other fingers flexed, wrist centred); Wrist Flexion (all

fingers neutral, MG996R rotates wrist downward 45 degrees); Wrist Extension (all fingers neutral, MG996R rotates wrist upward 45 degrees); Thumbs Up (thumb fully extended, all other fingers flexed, wrist centred -- also used as mode-selection toggle). All eight classes produce a distinct physical posture in the 5-finger hand; no class is purely a control command, since the tendon-driven hand can independently position each finger.

Safety logic prevents unintended actuation: a confidence threshold (initially 0.7) rejects low-confidence predictions; temporal debounce requires two consecutive identical classifications before issuing a command; and a command hold time of 500 ms prevents rapid servo oscillation. An emergency stop mechanism (physical button and BLE command) immediately de-energises the servos.

3.5 Ethics, Safety, and Data Considerations

Human-subject testing will be conducted only after institutional ethics approval and informed consent. The prototype is a research demonstrator, not a certified medical device. Participant data will be pseudonymised and stored with access restricted to the research team. The actuator system includes software limits and a physical power-disconnect mechanism for immediate test termination. Participants may withdraw at any time without penalty.

3.6 Cost Evaluation

Because "low-cost" is a central claim in the project title, cost is treated as a measurable design requirement. The USD 20 target refers specifically to Tier A: the electronics BOM (integrated circuits, passive components, and connectors). This is compared against published electronics costs: Molinari et al. report more than USD 500 [4]; commercial systems range from USD 3,000 to USD 8,000. The comparison demonstrates the cost reduction, not merely the absolute value.

Table 5: Cost Tiers

Tier	Contents	Target
A: Electronics BOM	ICs, passives, connectors	< USD 20 (approx. KES 2,600; current BOM KES 3,150 -- target via component optimisation)
B: Prototype BOM	Tier A + PCB fab, 3D printing,	< USD 65 (approx. KES 8,700;

	batteries, servos	current full BOM KES 8,700)
C: Volume (100+ units)	Tier A at volume pricing	< USD 15 (approx. KES 1,950 at volume pricing)

3.7 System Testing and Evaluation

Testing is organised into four layers corresponding to the system modules:

Table 4: Performance Targets

Layer	Metric	Target
Electronics	Signal-to-noise ratio	> 20 dB
Electronics	Inter-channel crosstalk	< -40 dB
Electronics	50 Hz notch rejection	> 30 dB
ML (offline)	Macro F1 on NinaPro DB5	≥ 0.85
Embedded ML	INT8 model size	< 100 KB
Real-time	Model inference latency	< 50 ms
System	Mean accuracy across 5 subjects	$\geq 80\%$
Cost	Electronics BOM (Tier A)	< USD 20
Cost	Prototype BOM (Tier B)	< USD 65

3.8 End-to-End Demonstration

The final demonstration shows the system operating without a USB or PC connection:

1. A participant is fitted with four Ag/AgCl surface electrodes on the forearm using the placement defined in the signal specification.
2. The BLE application confirms device connectivity and guides the participant through the approximately three-minute calibration sequence.
3. The ESP32-S3 acquires the four channels at 2 kHz, performs digital filtering, extracts the 24-element feature vector, and classifies the gesture.
4. The classified command is filtered by the confidence threshold and debounce logic, then translated into a servo position command via PCA9685.

5. The 5-finger tendon-driven 3D-printed prosthetic hand produces the finger-and-wrist posture corresponding to the classified gesture.
6. The BLE mobile application displays the predicted gesture, confidence score, and live channel activity while performance data are logged for analysis.
7. The entire system is powered from 9 V batteries with no external connections.

CHAPTER FOUR

EXPECTED RESULTS

This chapter states the technical results expected from the proposed work, linked directly to the three specific objectives. At proposal stage, all results are stated as targets rather than achieved outcomes.

4.1 Expected Electrical Performance (Objective 1: Design)

The four-channel acquisition board is expected to achieve a signal-to-noise ratio above 20 dB when measuring a 1 mV peak-to-peak sEMG-equivalent test signal at 100 Hz. The frequency response is expected to show -3 dB points at approximately 20 Hz and 500 Hz with a flat passband (variation less than 3 dB). The 50 Hz notch filter is expected to achieve rejection exceeding 30 dB. Inter-channel crosstalk is expected to remain below -40 dB across all four channels. These targets are informed by simulation results (Sallen-Key standalone: -3 dB at 22 Hz and 492 Hz; twin-T notch depth: 32.3 dB) and will be validated through laboratory measurements on the fabricated PCB.

4.2 Expected Embedded Performance (Objective 2: Implementation)

The ESP32-S3 is expected to maintain stable four-channel sampling at 2 kHz (8 kHz aggregate) via DMA without sample loss. Feature extraction (24 time-domain features from a 500-sample window) is expected to complete within 5 ms, well within the 50 ms step interval. The INT8-quantized 1D-CNN model is expected to occupy less than 100 KB of flash, use less than 50 KB of SRAM for inference, and complete inference within 50 ms. Peak RAM usage for the complete pipeline (sampling buffer, feature extraction, inference) is expected to remain below 200 KB, within the ESP32-S3's 512 KB SRAM.

4.3 Expected Classification Performance (Objective 2: Implementation)

The 1D-CNN is expected to achieve an offline macro F1 score of 0.85 or higher on held-out NinaPro DB5 test data with subject-independent splitting. This target is conservative relative to the 97 percent accuracy reported by Mekruksavanich et al. [2] on the same dataset, accounting for the reduced channel count (4 versus 16), feature-based rather than raw-signal input, and INT8 quantization. Baseline classifiers

(LDA, SVM, RF) are expected to achieve F1 scores between 0.70 and 0.80, establishing the performance reference.

4.4 Expected Calibration Improvement (Objective 2: Implementation)

User-specific calibration via transfer learning is expected to improve cross-user recognition accuracy by at least 10 percentage points over the generic pre-trained model. This expectation is based on findings by Lee et al. [3], who demonstrated that fine-tuning only the last classification layer with approximately two minutes of user data is sufficient for meaningful personalisation.

4.5 Expected End-to-End System Performance (Objective 3: Test and Evaluate)

The complete system, validated with five test subjects, is expected to achieve a mean real-time accuracy of 80 percent or higher across all eight gesture classes. Actuation correctness -- the proportion of classified gestures that produce the intended finger-and-wrist posture on the prosthetic hand -- is expected to be within 5 percentage points of classification accuracy, with the difference attributable to the confidence filtering that rejects uncertain predictions. Total system response time (from end of analysis window to servo motion onset) is expected to be approximately 300 ms (50 ms inference + decision filtering + servo latency).

With five subjects, the study provides an initial feasibility demonstration. Statistical claims will be limited to descriptive statistics (mean, standard deviation, range) and per-subject results; inferential statistical tests require larger sample sizes. This limitation will be stated explicitly in the final report.

4.6 Expected Cost Outcome (Objective 3: Test and Evaluate)

The electronics BOM (Tier A) is expected to cost less than USD 20 (approximately KES 2,600). The complete prototype BOM (Tier B), including PCB fabrication, 3D-printed prosthetic hand, batteries, and servo motors, is expected to cost less than USD 65 (approximately KES 8,700). These figures will be reported alongside published costs from the literature (Molinari et al.: more than USD 500; commercial: USD 3,000 to USD 8,000) to demonstrate the cost reduction quantitatively.

PROJECT BUDGET

Table 6: Project Budget

Item	Description	Qty	Rate (KES)	Amount (KES)
1	ESP32-S3-DevKitC-1	1	800	800
2	INA128 Instrumentation Amplifier	4	350	1,400
3	MCP6002 Rail-to-Rail Op-Amp	8	50	400
4	Passive components (resistors, capacitors)	1 set	300	300
5	Ag/AgCl Surface Electrodes (50 pack)	1	500	500
6	PCA9685 I2C PWM Driver	1	250	250
7	SG90 Micro-Servo Motors (fingers x5)	5	150	750
8	MG996R Servo Motor (wrist)	1	400	400
9	3D Printing (5-finger prosthetic hand, PETG filament)	1	1,500	1,500
10	4-Layer PCB (JLCPCB, incl. shipping)	1	1,200	1,200
11	9V Batteries + Holders	2	150	300
12	Connectors, headers, wires, enclosure	1 set	500	500
13	Electrode cables and shielded	4	100	400

	leads			
			TOTAL	KES 8,700

Note: The electronics BOM (Tier A: items 1-4, 6) totals KES 3,150 (approximately USD 24), which is close to the USD 20 target and will be optimised through component substitution and volume procurement. The complete prototype BOM (Tier B: all items) totals KES 8,700 (approximately USD 67), which slightly exceeds the USD 65 target due to the upgrade from a 2-servo gripper to a 6-servo 5-finger prosthetic hand; the Tier B target will be reviewed as component prices are confirmed.

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